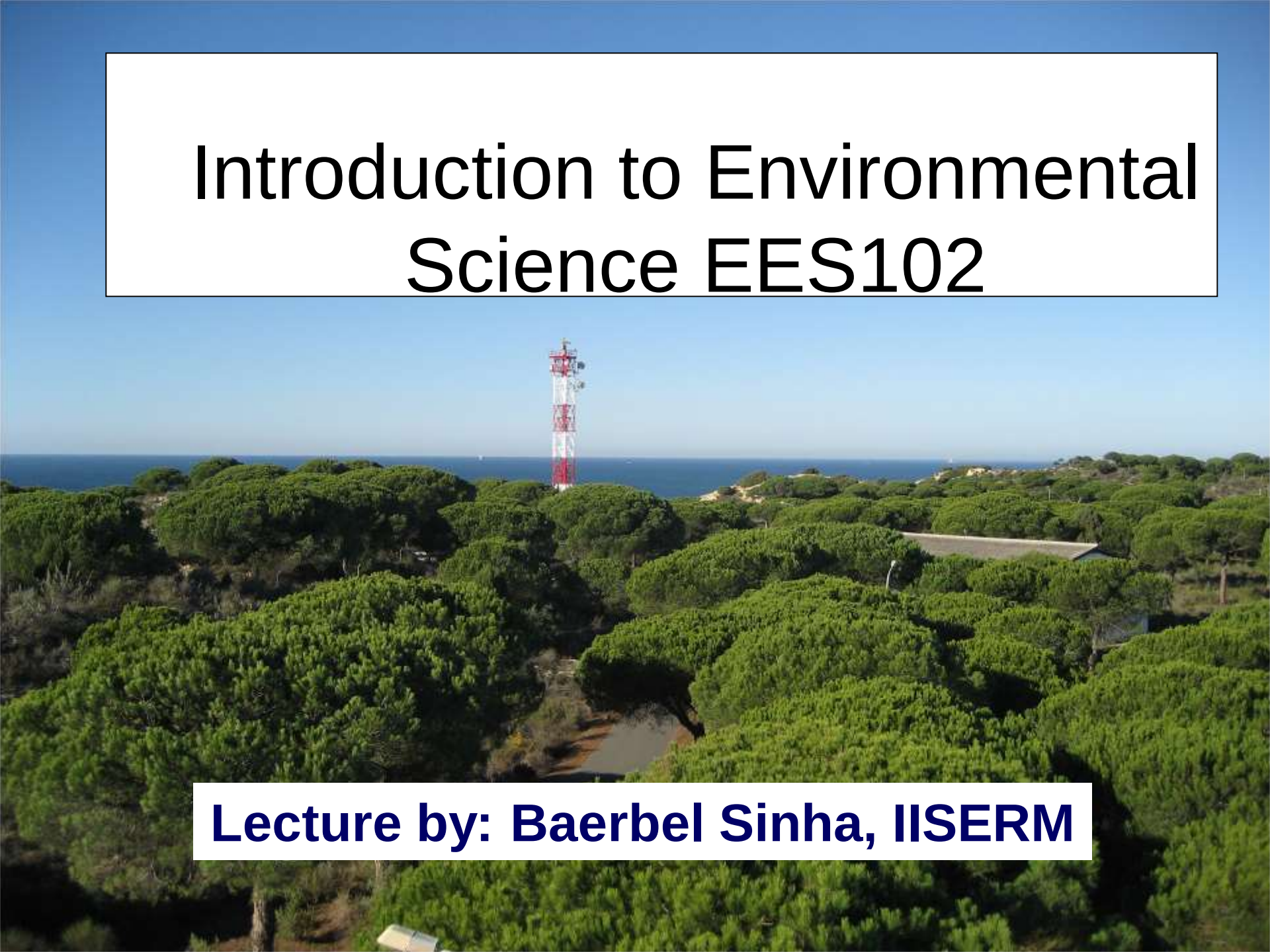


Introduction to Environmental Science EES102



Lecture by: Baerbel Sinha, IISERM

GLOBAL WARMING POTENTIAL – PART 1

Global warming potential (GWP) is the heat absorbed by any greenhouse gas in the atmosphere, as a multiple of the heat that would be absorbed by the same mass of carbon dioxide (CO₂) being emitted at the same point in time over the same period of time. GWP is 1 for CO₂

Different greenhouse gasses such as CO₂, CH₄, N₂O, CFCs and SF₆ absorb different wavelength ranges of the terrestrial radiation. Depending on how much or how little of the terrestrial radiation is at the right wavelength and how much of that wavelength is already absorbed by water vapour, their power to change global temperatures varies.

Different gasses have different lifetime. Some stick around relatively shortly e.g. methane (12.4 years) others for longer time periods e.g. CO₂ ~95 years, SF₆ 3200 years.

GLOBAL WARMING POTENTIAL – PART 2

The GWP it depends on the gas and the time frame of consideration.

Since some gases, again like methane, break down much faster than CO₂. They have a shorter atmospheric lifetime.

As a consequence their heat absorption, or GWP, over the next 20 years is a bigger multiple of CO₂ (83) than their heat absorption over 100 years will be (30) years.

So in short when we calculate the GWP we look at 1 ton of CH₄ emitted now and ask how many tonnes of CO₂ would have to be emitted to do the same damage to the climate system as 1 ton of CH₄ does in 20 years (Answer: 96 tonnes of CO₂).

G.W.P OF DIFFERENT GHGs RELATIVE TO CO₂

Gas name	Chemical formula	Lifetime (years) [11]: Table 7.15 [13]	Efficiency (Wm ⁻² ppb ⁻¹ , molar basis). [11]: Table 7.15 [13]	20 year GWP[11]: Table 7.15 [13]	100 year GWP[11]: Table 7.15 [13]	500 year GWP[11]: Table 7.15 [14]
Carbon dioxide	CO ₂	(A)	1.37×10^{-5}	1	1	1
Methane (fossil natural gas)	CH ₄	12	5.7×10^{-4}	83	30	10
Methane (pure non-fossil)	CH ₄	12	5.7×10^{-4}	81	27	7.3
Nitrous oxide	N ₂ O	109	3×10^{-3}	273	273	130
CFC-11 (R-11)	CCl ₃ F	52	0.29	8,321	6,226	2,093
CFC-12 (R-12)	CCl ₂ F ₂	100	0.32	10,800	10,200	5,200
HCFC-22 (R-22)	CHClF ₂	12	0.21	5,280	1,760	549
HFC-32 (R-32)	CH ₂ F ₂	5	0.11	2,693	771	220
HFC-134a (R-134a)	CH ₂ FCF ₃	14	0.17	4,144	1,526	436
Tetrafluoromethane (R-14)	CF ₄	50,000	0.09	5,301	7,380	10,587
Hexafluoroethane	C ₂ F ₆	10,000	0.25	8,210	11,100	18,200
Sulfur hexafluoride	SF ₆	3,200	0.57	17,500	23,500	32,600

RADIATIVE FORCING AND G.W.P.

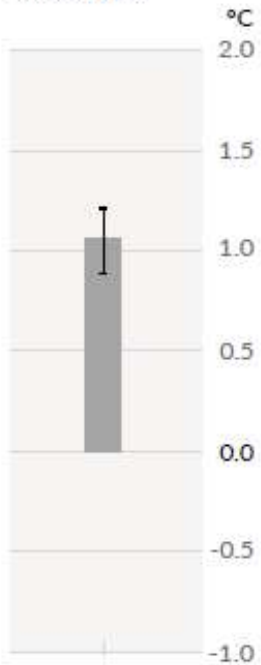
- By computing radiative forcings associated with changes in emissions of individual GHG, we can assess and compare the potential climate effects of different gases & make appropriate policy decisions
- International environmental policy on GHGs emissions makes use of the Global Warming Potential (G.W.P.) index, which is defined as the radiative forcing resulting from an instantaneous 1 kg injection of a given gas into the atmosphere relative to the radiative forcing from an Instantaneous 1 kg injection of CO₂:

$$GWP = \frac{\int_{t_0}^{t_0 + \Delta t} \Delta F_{1kg \text{ X}} dt}{\int_{t_0}^{t_0 + \Delta t} \Delta F_{1kg \text{ CO}_2} dt}$$

WHAT IS THE CURRENT STATUS OF WARMING?

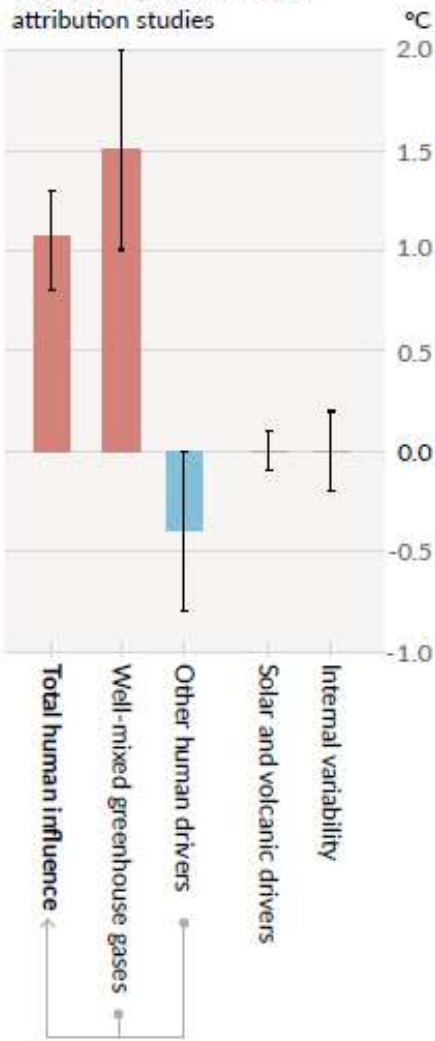
Observed warming

a) Observed warming 2010-2019 relative to 1850-1900

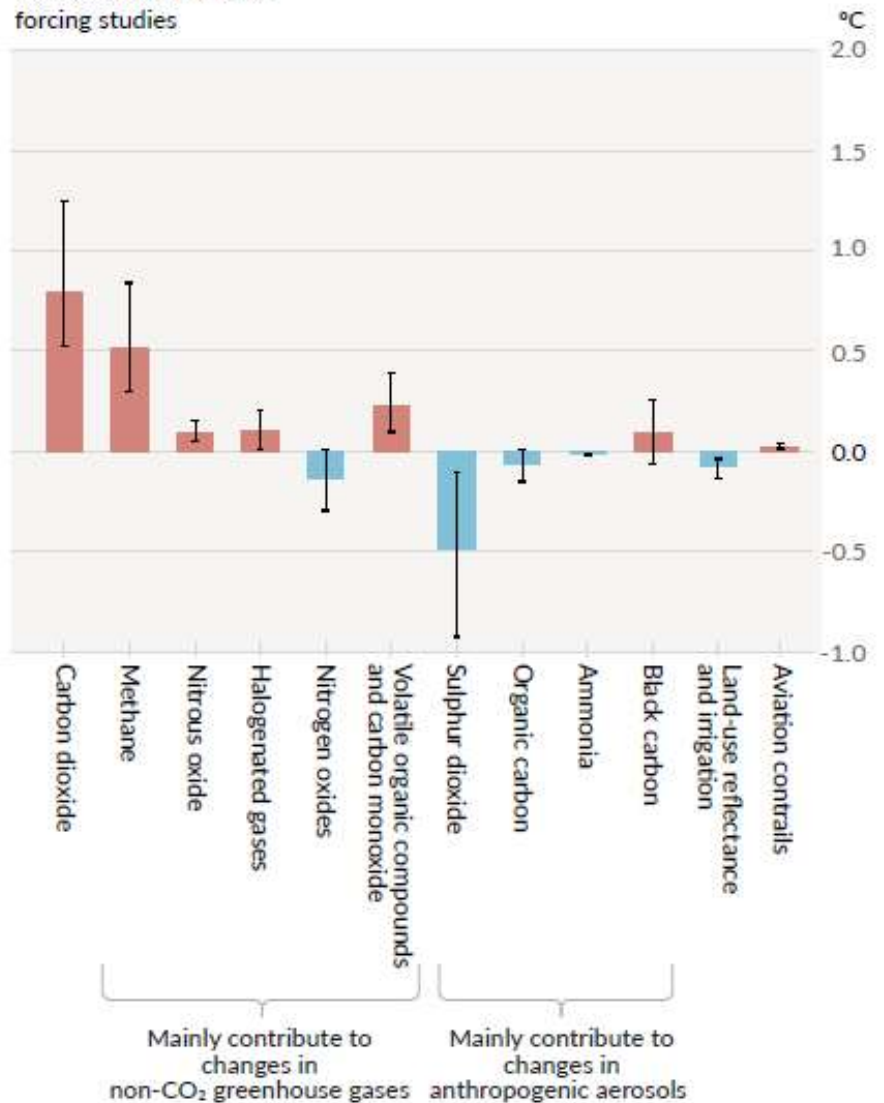


Contributions to warming based on two complementary approaches

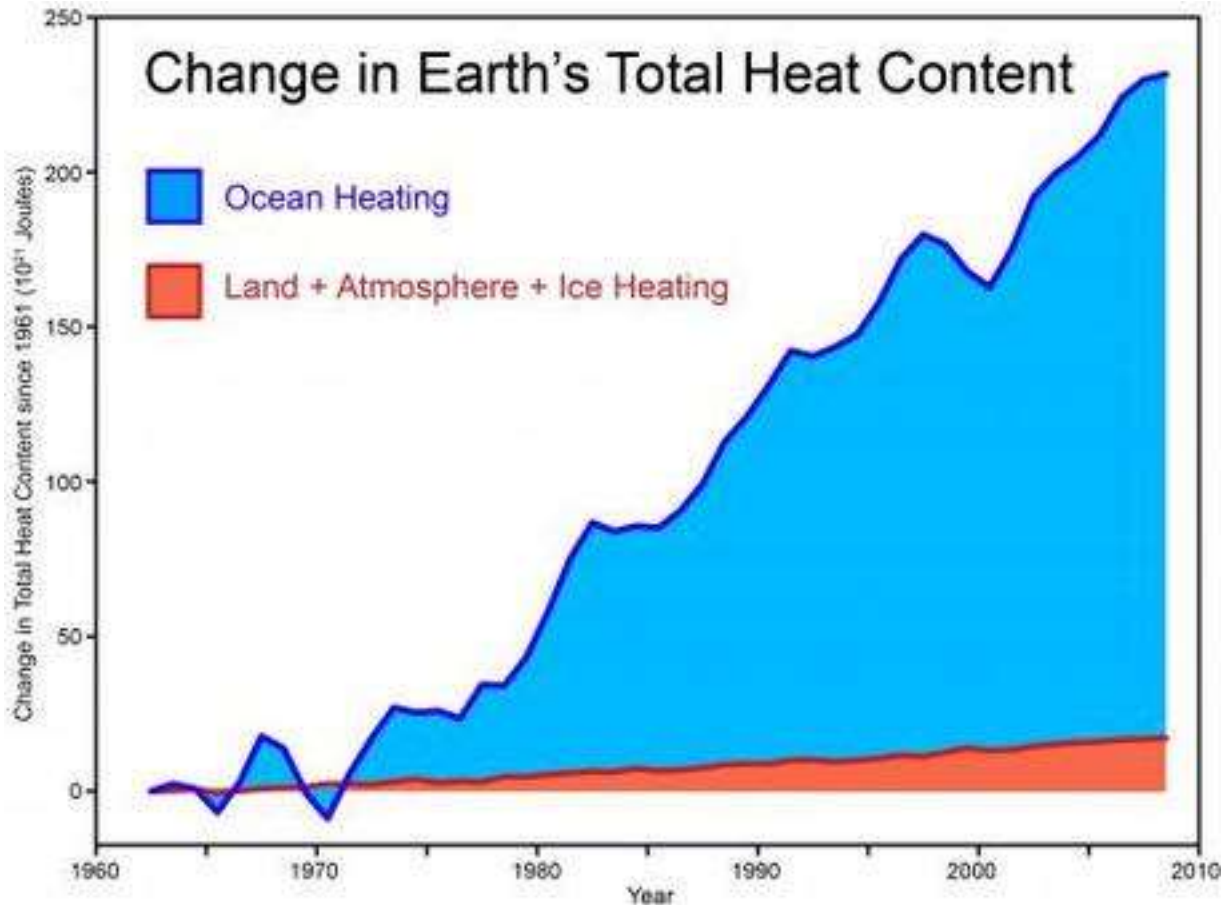
b) Aggregated contributions to 2010-2019 warming relative to 1850-1900, assessed from attribution studies



c) Contributions to 2010-2019 warming relative to 1850-1900, assessed from radiative forcing studies



WHERE HAS THE ADDITIONAL ENERGY GONE



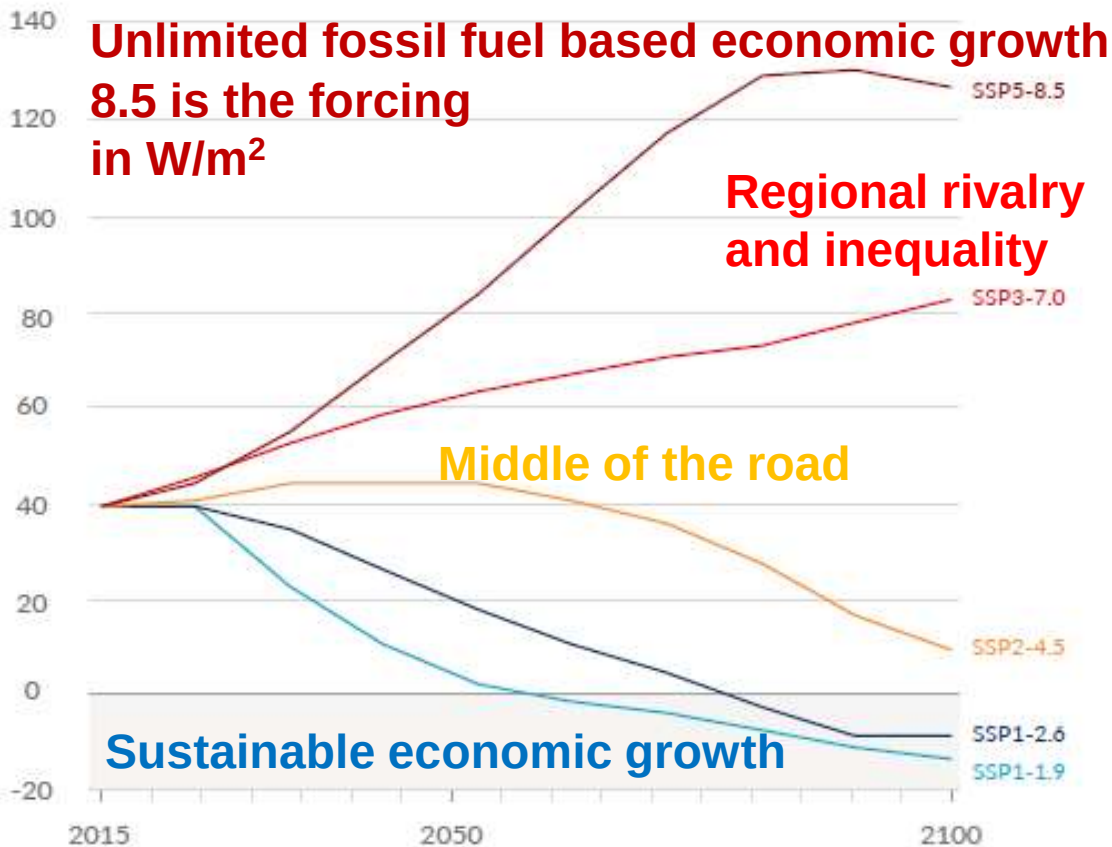
90% has gone into the ocean while 10% has been accommodated on the surface in the form of atmospheric heating, land heating and ice melting. Over time this change in ocean energy will result in thermal expansion of ocean water and the ice melting in additional water in the ocean. Both of these elements contribute to sea level rise. This heating will also reduce the amount of CO_2 the ocean can store

WHAT CLIMATE CHANGE SCENARIOS DOES THE IPCC MODEL?

Future emissions cause future additional warming, with total warming dominated by past and future CO₂ emissions

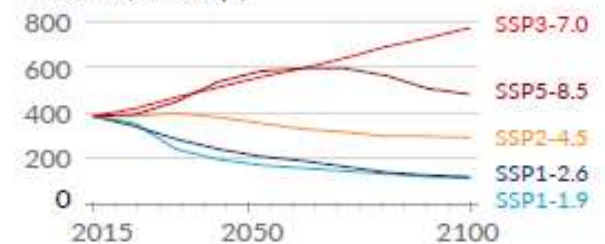
a) Future annual emissions of CO₂ (left) and of a subset of key non-CO₂ drivers (right), across five illustrative scenarios

Carbon dioxide (GtCO₂/yr)

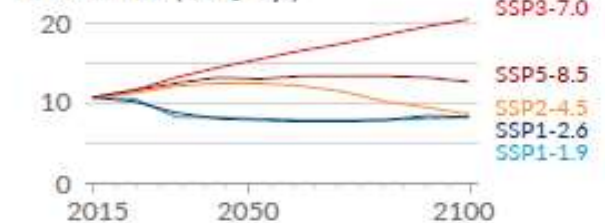


Selected contributors to non-CO₂ GHGs

Methane (MtCH₄/yr)

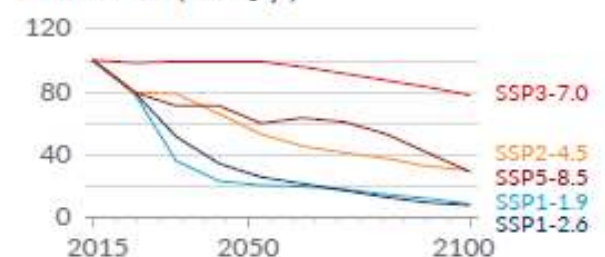


Nitrous oxide (MtN₂O/yr)



One air pollutant and contributor to aerosols

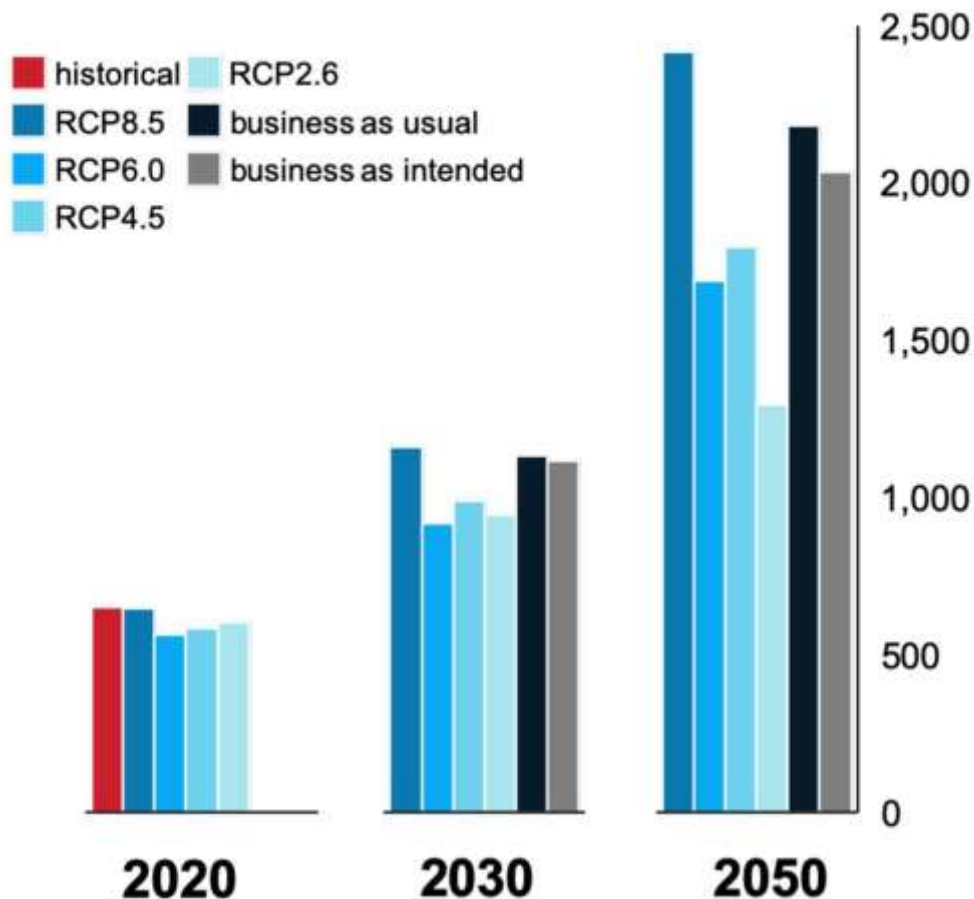
Sulfur dioxide (MtSO₂/yr)



WHICH ROUTE ARE WE ON?

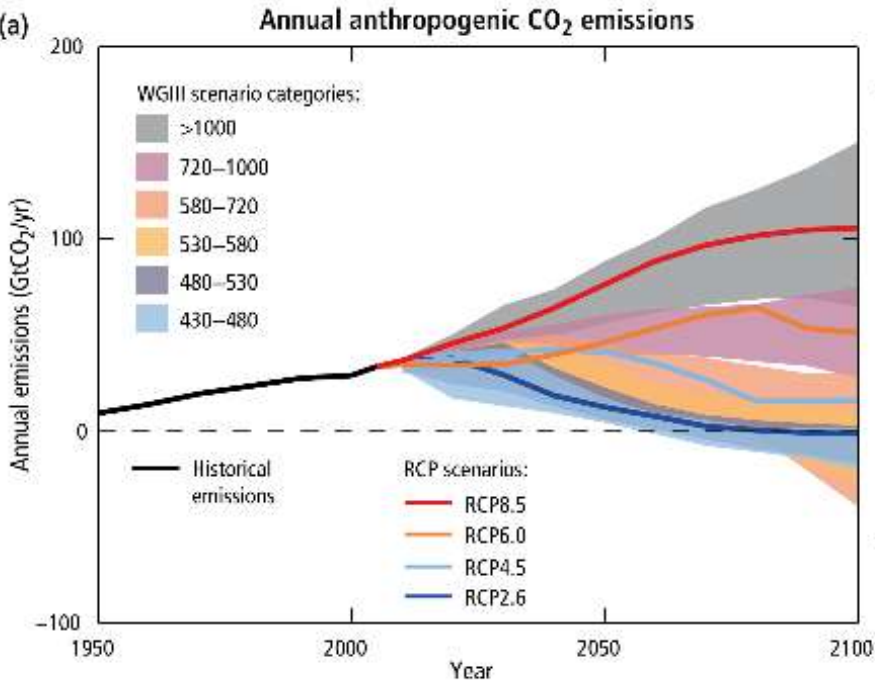
Cumulative emissions since 2005

Gt CO₂



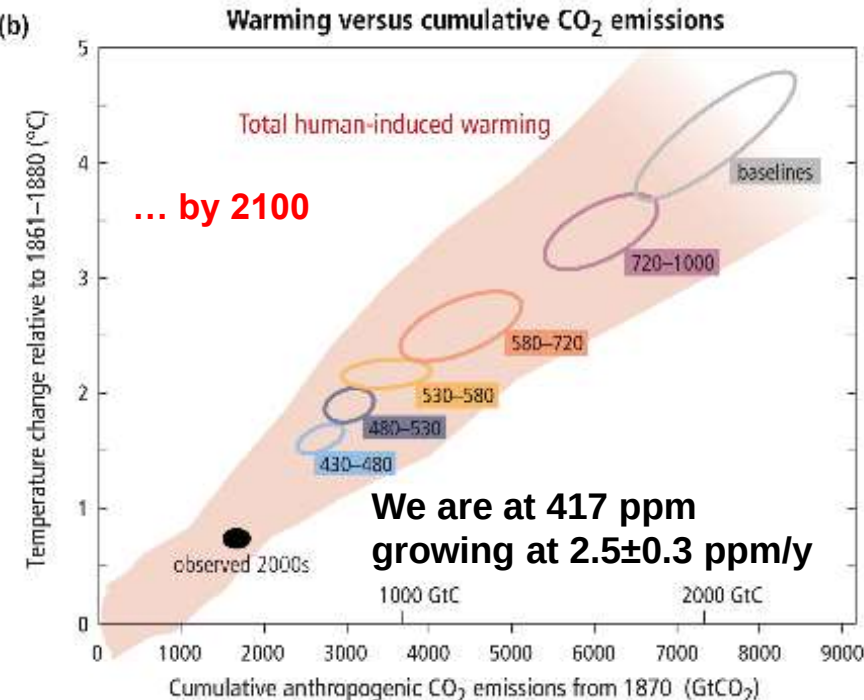
Schwalm et al. 2020
19656–19657 | PNAS |
August 18, 2020 |

Still on the worst track possible



CLIMATE CHANGE IN THE 5TH ASSESSMENT WAS MODELLED ACCORDING TO RCP “REPRESENTATIVE CONCENTRATION PATHWAYS” THE NUMBER STANDS FOR THE RADIATIVE FORCING +2.6, +4.5, +6.0, AND +8.5 W/M² RELATIVE TO PREINDUSTRIAL VALUES ATTAINED BY 2100

For RCP2.6 short term warming by 2100 will be +1.5 to +2 °C. Sustained net negative anthropogenic GHG emissions after the year 2070 are predicted. Global warming will reduce to 0.0 to 1.2 °C by the late-23rd century (2281-2300 average)



- **For RCP8.5 short term warming by 2100 will be +4 to +5 °C . For the extended RCP8.5, global warming of 3.0 to 12.6 °C is projected by the late-23rd century**

=> the uncertainty of the range steams from the fact that we do not know how fast the methane hydrates will melt. This frozen methane ice is on the ocean floor and its melting would catapult the planet into a “runaway greenhouse” which would take ~170,000 years to return to normal through natural processes

#ALLFORCLIMATE: SIX DEGREES

We are facing a climate emergency posing an existential threat to humanity. The crisis requires "rapid, far-reaching and unprecedented changes in all aspects of society", says the UN's IPCC*. If we want to ensure good living conditions for all, within the limits of the planet, we must redefine what living well means, and create an economy that does not depend on growth.

We have been waiting too long for others to solve the crisis. But all actors in society have important roles to play in the transition ahead. All of us, individuals and organisations, must realise that we are part of the problem and must act boldly now to become part of the solution. This is the core of the #AllForClimate campaign.

Despite the progress made by science, we do not know for sure when tipping points will be triggered, and when global warming could be unstoppable. For any tenth of a degree of global warming we let happen, we are gambling with a future that, in the worst case, none of us can survive.

** Intergovernmental Panel on Climate Change*



+0°C

THE WORLD BEFORE INDUSTRIALISATION

The temperature reached on Earth around 11,000 years ago after the last ice age. Ice melting made the sea level rise about 120 metres. Stable and mild climate allowed humans to grow food and settle down. Cities arose, consumption was low and living ways mostly sustainable. Humans were already affecting their environment, but not so much the world's climate.

+1°C

EVERY TENTH OF DEGREE OF WARMING COUNTS

In 2015, global warming reached 1°C above pre-industrial levels. Six years later it reached 1.1°C. The Arctic and Antarctica are melting. The Gulf Stream is weakening. Glaciers are melting affecting freshwater supply. Hurricanes, droughts, heatwaves and large wildfires are more frequent. Low-lying coastlines are flooded. Ice-free sea absorbs more heat and accelerates global warming. Corals are bleaching. Wildlife is in decline. Food production is starting to be affected.

+2°C

A PRESCRIPTION FOR LONG-TERM DISASTER

Climate breakdown. Ice-free summertime in the Arctic. Irreversible loss of the West Antarctic Ice Sheet. 80 million people displaced due to sea-level rise. Deadly dengue fever. About half a million die of malnutrition. Insect crop pests. Hundreds of millions exposed to deadly heat. Shortage of water in Africa. Mass migration. Loss of glaciers affects freshwater supply. More floods. Conflagration of the Amazon rainforest. Many species are threatened. 99% of reef-building corals are lost.

+3°C

VEGETATION EMIT CARBON AND SPEED WARMING

Ice sheets and glaciers partly collapse. Sea-level rises to 8-14 metres. Extreme high tides, floods, superstorms and super-hurricanes. Hundreds of millions of people are forced to relocate. Half the world's population is exposed to deadly heat waves. Invading deserts. Food production cut by half. Mass starvation. Death of the Amazon rainforest. Wildlife disruption. Carbon release from vegetation and soils speeds global warming, so does methane emissions from melting permafrost.

+4°C

GLOBAL WARMING IS UNSTOPPABLE

Three-quarters of the global population is exposed to deadly heat in days. Substantial areas of the planet are biologically uninhabitable for humans. Climate refugee numbers become unmanageable. Half the world's land surface becomes 'arid'. Extreme droughts and floods. Category 6 super-hurricanes. Synchronous failures of major world food crops. Potential ice-free planetary state. Mass extinction. Permafrost meltdown makes the escalation of global warming unstoppable.

+5°C

SURVIVING HUMANS LIVE LIKE ANIMALS

Ice sheet is permanently gone. Humans can no longer survive outside for a long time. Food production is decimated. Complex human societies have collapsed. Global famine. Surviving humans try vainly to live like animals off the land.

+6°C

ONLY FEW ANIMALS AND FUNGI SURVIVE

Extreme greenhouse effect destroys most of life on Earth. Only a few animals can survive. Ecosystems and food webs cease to exist in any functional sense. The big winners are detritivores, bacteria and fungi.

The descriptions on this poster are mainly based on the book "Our Final warning: six degrees of climate emergency" written by award winning Journalist Mark Lynas and published in 2020.

Read more at www.allforclimate.today #ClimateForAll #AllForClimate



WHAT WILL CLIMATE CHANGE BRING?

Hot & dry areas will mostly become hotter and dryer. Their biomes will adjust to less vegetation via wildfires.

Hot and wet areas will mostly become hotter and wetter. A warmer atmosphere can hold more water which will trigger more extreme rain events and floods. Some areas in the tropical zone may become uninhabitable because the wet bulb temperatures will be too high for land-based life.

The arctic and high altitudes will change most rapidly. Their melting permafrost and warming peatlands will accelerate global change. Fires will become common

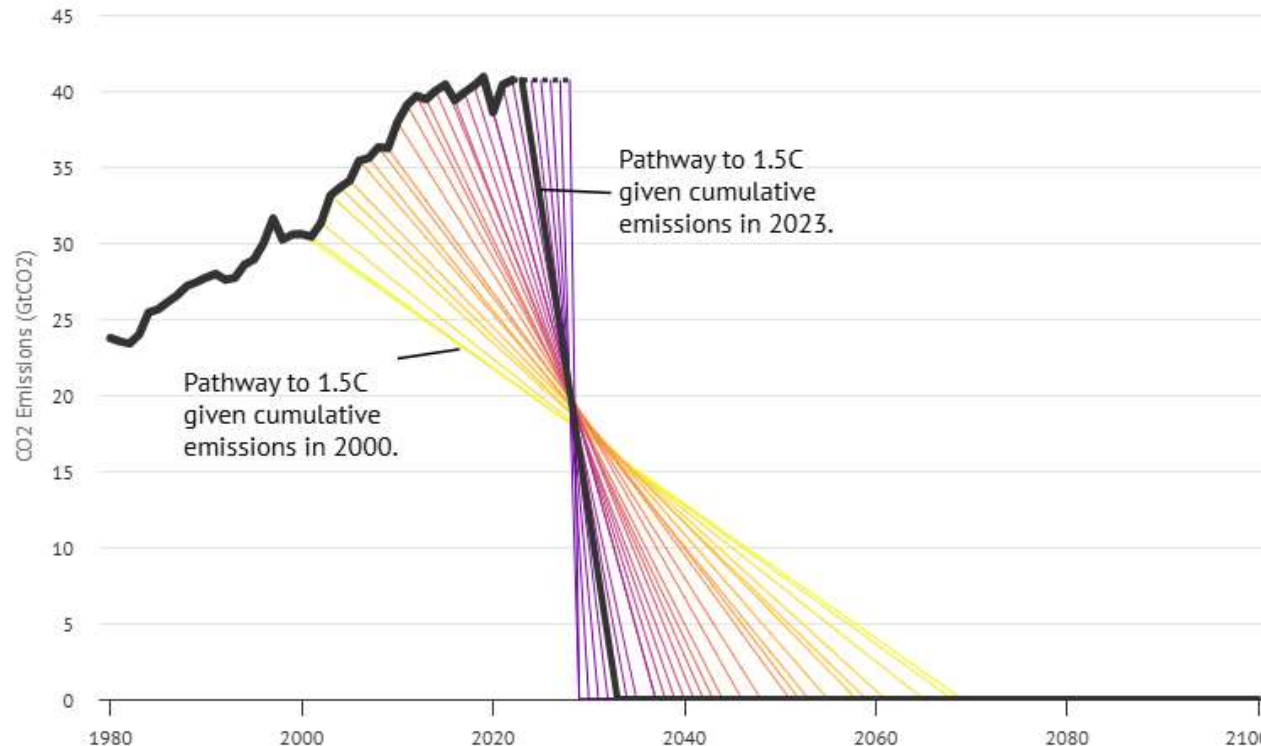
Vector borne/tropical diseases will spread to new areas

Global crop yields will suffer from heat stress and reducing water availability

Sea level rise will affect millions directly or indirectly (via saltwater intrusion into coastal aquifers)

HOW URGENT COMBATTING CLIMATE CHANGE?

Limiting warming to 1.5C is virtually impossible without net-negative emissions



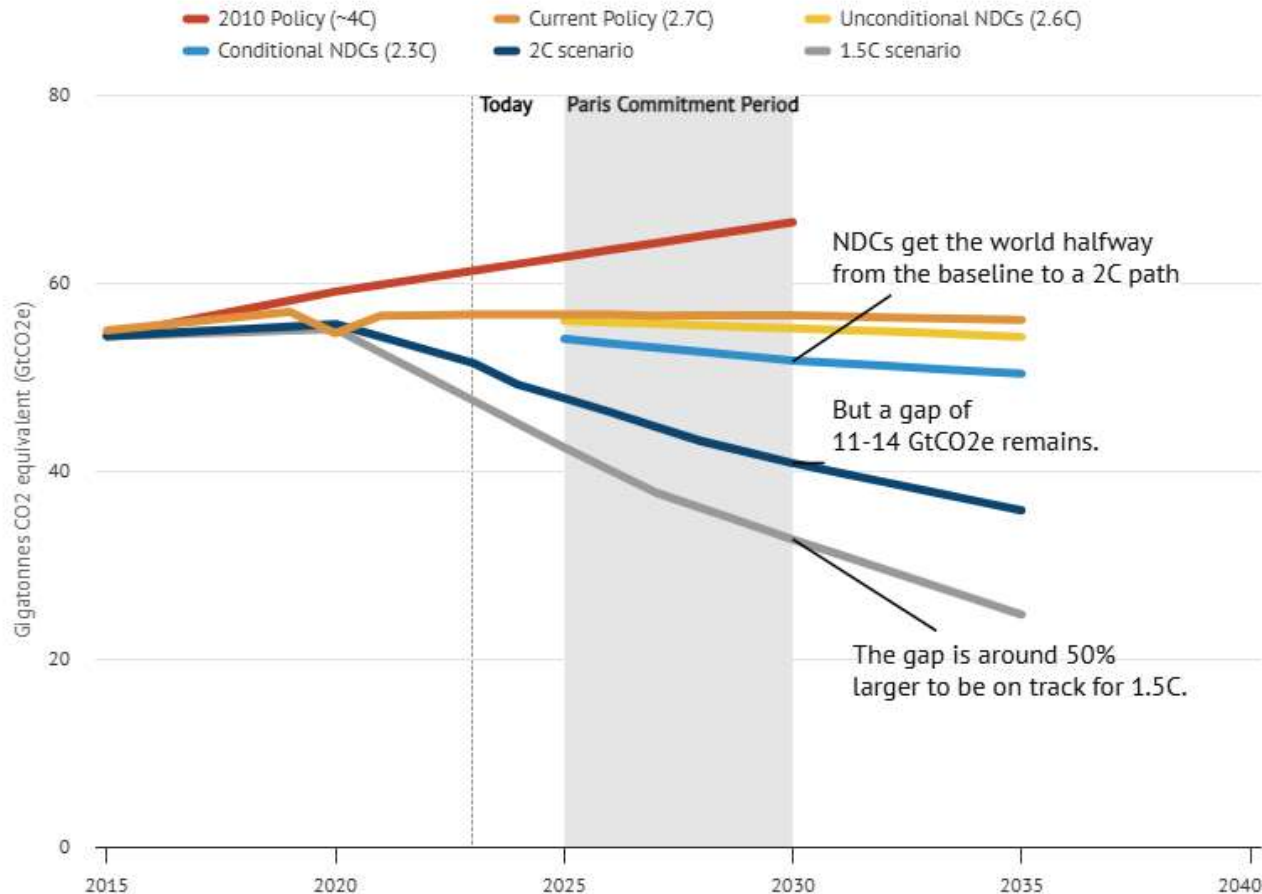
Source: UNEP Gap report.

CarbonBrief
CLEAR ON CLIMATE

Emission reduction trajectories associated with a 50% chance of limiting warming below 1.5C, without a reliance on net-negative emissions, by starting year. Solid black line shows historical emissions, while dashed black line shows emissions constant at 2023 levels. Source: Historical CO2 emissions from the Global Carbon Project. 1.5C carbon budgets based on [Lamboll et al 2023](#). Chart by Carbon Brief, adapted from a figure originally designed by [Robbie Andrew](#).

THE AMBITIONS GAP

Greenhouse emissions remain far off track for global climate goals



Source: UNEP Gap report.

CarbonBrief
CLEAR ON CLIMATE

Median emission scenarios adapted from Figure 4.2 in the 2023 UNEP Emission Gap Report. Red line shows a scenario with no new climate policies after 2010, orange shows existing policies already implemented by governments, yellow and light blue lines show additional conditional and unconditional NDCs, respectively. The dark blue line shows emissions consistent with a below 2C trajectory, and grey line shows emissions consistent with a 1.5C trajectory. Chart by Carbon Brief.

WHAT DOES “NET ZERO” MEAN

Net zero refers to a state in which the greenhouse gases going into the atmosphere are balanced by removal out of the atmosphere.

The term net zero is important because – for CO₂ at least – this is the state at which global warming stops. The Paris Agreement underlines the need for net zero. It *requires* states to ‘achieve a balance between anthropogenic emissions by sources and removals by sinks of greenhouse gases in the second half of this century’.

PROBLEMS WITH SOCIETAL RELEVANCE

- ⊙ climate change – greenhouse effect
 - ⊙ London Smog
 - ⊙ acid rain, water pollution
 - ⊙ Los Angeles Smog (ozone smog)
 - ⊙ health effects from aerosols
 - ⊙ stratospheric ozone hole
 - ⊙ self-cleansing capacity of the atmosphere (free radicals)
- Trajedy of commons
- } air quality
NIMBY
- Recovering thanks to Montreal protocol

GEOENGINEERING AS AN ALTERNATIVE TO NET ZERO

Geoengineering is the deliberate large-scale interventions in the Earth's climate system intended to counteract human-caused climate change.

The term commonly encompasses two broad categories: large-scale carbon dioxide removal (CDR) and solar radiation modification (SRM).

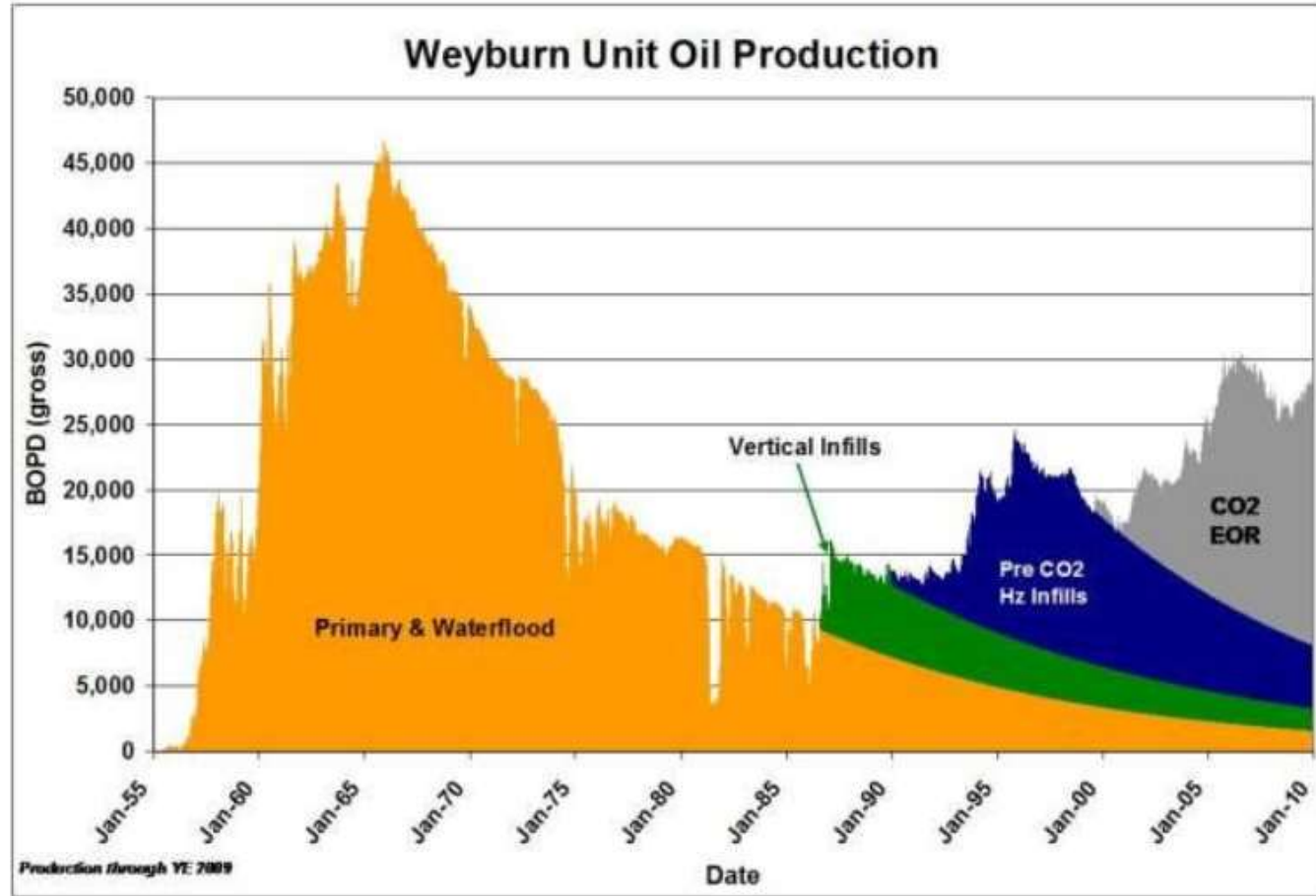
CDR involves techniques to remove carbon dioxide from the atmosphere and is generally considered a form of climate change mitigation.

today's global carbon dioxide removal (CDR) capacity amounts to only about 0.01 GT of CO₂ per year across all nature- and engineered-based pathways combined and occurs mainly in the form of EOR

SRM aims to reduce global warming by reflecting a small portion of sunlight (solar radiation) away from Earth and back into space

Radiative cooling is a promising form of SRM

ENHANCED OIL RECOVERY

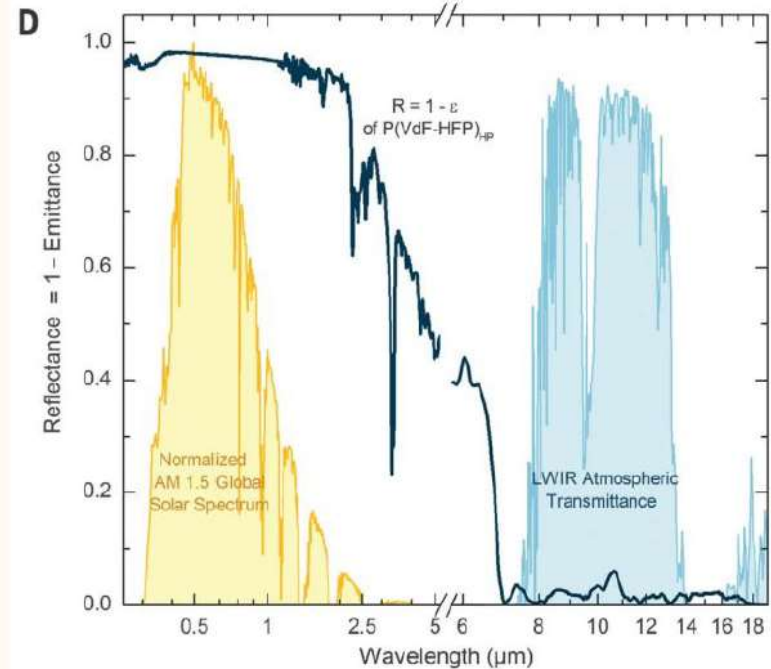


Is a new strategy of “greenwashing” where taxpayers pay oil and gas companies to capture CO2 from flue gas and inject it into oil fields to extract more oil and make more profit. Currently 90% of the global CDR capacity is EOR capacity. Only one project in Norway stores CO2 in a saline aquifer purely to remove it from the atmosphere.

RADIATIVE COOLING

General Characteristics

1. The reflectance of the system is maximum in the region where the solar irradiance is maximum (orange region).
2. The emittance ($1 - \text{Reflectance}$) of the system is maximum in the Atmospheric IR Window (blue region).



Hierarchically porous polymer coatings for highly efficient passive daytime radiative cooling. *Science*, Mandal *et.al.*

Imaging a material that is a perfect mirror in the visible (Albedo 1) and emits all heat energy in the atmospheric window between 8-13 μm. Such a material could use space as a heat sink for cooling without using electricity.

RADIATIVE COOLING

	Steady state temperature below ambient temperature	Cooling power	Reflectivity in shortwaves	Emissivity in atmospheric window
HfO ₂ + SiO ₂ solid layer (Raman <i>et.al.</i> , 2014)	4.9°C ± 0.15°C	40.1 ± 4.1 Wm ⁻²	0.97	0.5-0.8
P(VdF-HFP) _{HP} paint (Mandal <i>et. al.</i>)	6°C	96 Wm ⁻²	0.96	0.97
Silver coated with PDMS (Kou <i>et.al.</i>)	8.2°C	127 Wm ⁻²	0.9	0.9-1.0

Diverse approaches and lot of new potentially valuable patents.

Spacecool Inc. (Japan) <https://spacecool.jp/en/product/>

SkyCool Systems Inc. (USA) <https://www.skycoolsystems.com/>

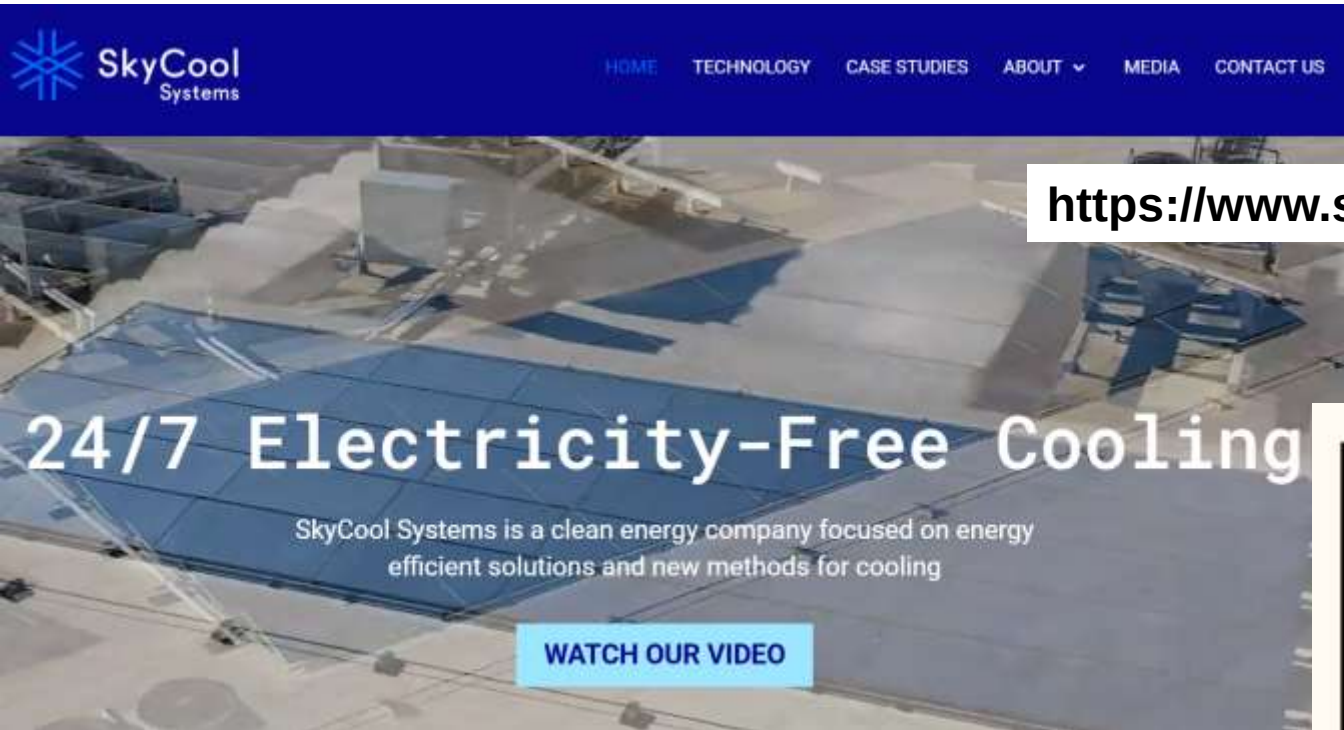
Radi-Cool Sdn Bhd (Malaysia) <https://www.radi-cool.com.my/>

ChillSkyn 3M (Canada) <https://www.chillskyn.com/>

Cable Coatings Limited (USA) <https://www.assetcool.com/>

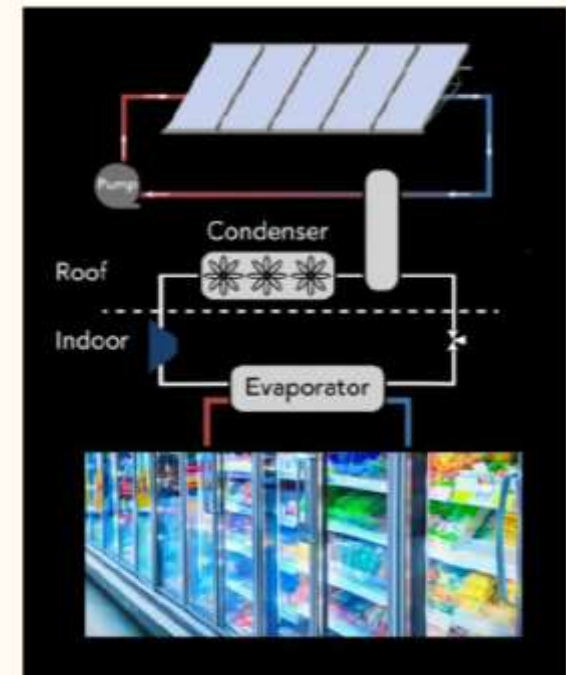
RADIATIVE COOLING

Some technology has been commercialized



<https://www.skycoolsystems.com/>

The next stage of Ujala needs to be for SkyCool type systems. India needs a competitive domestic manufacturing chain for affordable electricity free cooling systems!

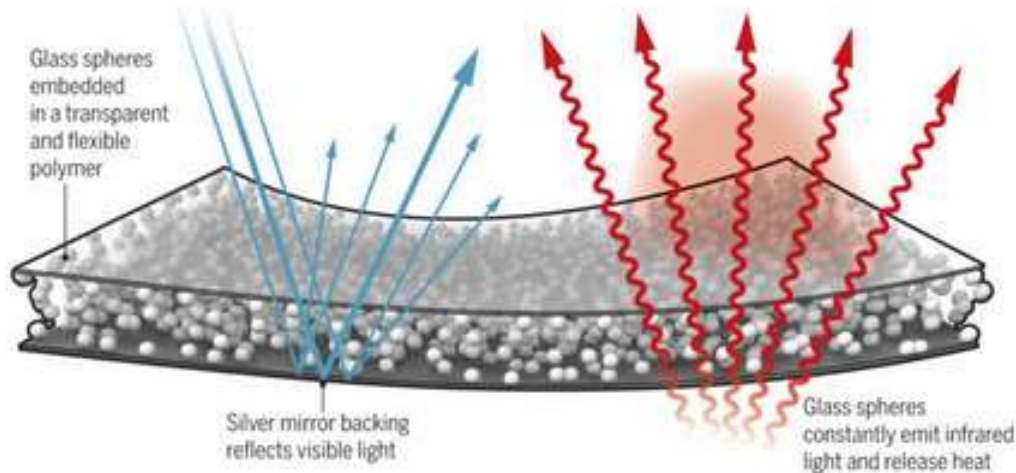


RADIATIVE COOLING

Some technology has been commercialized

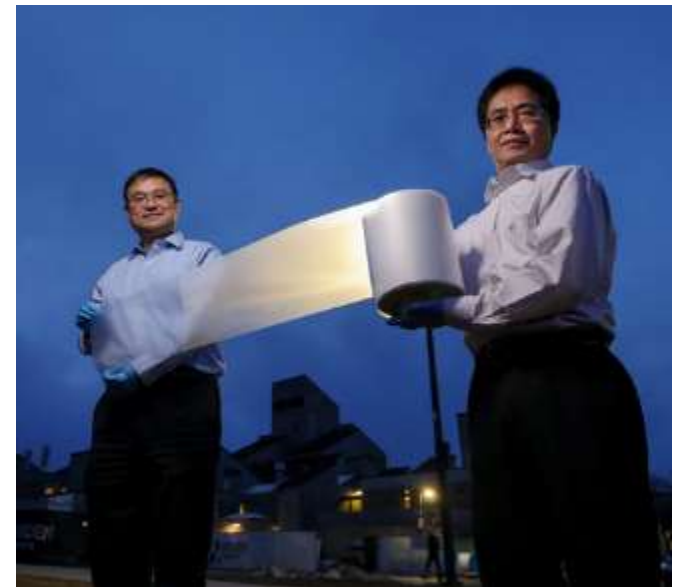
A flexible route to coolness

In previous approaches, intricate crystalline nanostructures emitted thermal infrared light. Zhai et al. use larger glass spheres ($\sim 10\ \mu\text{m}$ diameter) in a flexible polymer to create a scalable, thin-film cooling material.



Radiative cooling is the natural process through which objects shed heat in the form of infrared (IR) radiation. All materials at room temperature emit infrared at wavelengths of $5\text{--}15\ \mu\text{m}$. However, the process is typically not very efficient because it is counteracted by external influences that heat the object, such as sunlight and air currents. Air, meanwhile, absorbs and emits very little radiation with wavelengths $8\text{--}13\ \mu\text{m}$. The Earth cools itself at night by emitting infrared through this "atmospheric window" into space.

<https://radi-cool.com/#>



The next stage of Ujala needs to be for SkyCool type systems. India needs a competitive domestic manufacturing chain for affordable electricity free cooling systems!

RADIATIVE COOLING IS GEO-ENGINEERING !



Something that might happen for free but concentrated in urban areas

An antidote to the urban heat island

(B) Net radiation (incoming from the sun and outgoing from the Earth) for 2017 from monthly averages.³ If an area of the Earth equivalent to the regions depicted by white rectangles (total area of $5 \times 10^{12} \text{ m}^2$) were covered by currently available radiatively emitting surfaces, the total global heat flux could be reduced by nearly 1 W/m^2 , potentially enough to stop the increase of global temperatures.

SHORT LIVED CLIMATE FORCERS

SLCP

Short-lived climate pollutants are powerful climate forcers that remain in the atmosphere for a much shorter period of time than carbon dioxide (CO₂), yet their potential to warm the atmosphere can be many times greater.

The short-lived climate pollutants black carbon, methane, tropospheric ozone are the most important contributors to the man-made global greenhouse effect after carbon dioxide, responsible for up to 45% of current global warming. If no action to reduce emissions of these pollutants is taken in the coming decades, they are expected to account for as much as half of warming caused by human activity.

However if we eliminate their emission, where feasible and financially attractive, the effect in terms of reduction of radiative forcing is almost immediate.

Mitigating SLCP can help us avoid catastrophic climate change.

MITIGATION OF GASES WITH A SHORT LIFETIME CAN BUY US MOST TIME

Tropospheric ozone has a radiative forcing of $\sim 0.35 \text{ W/m}^2$ and is continuously produced and destroyed.

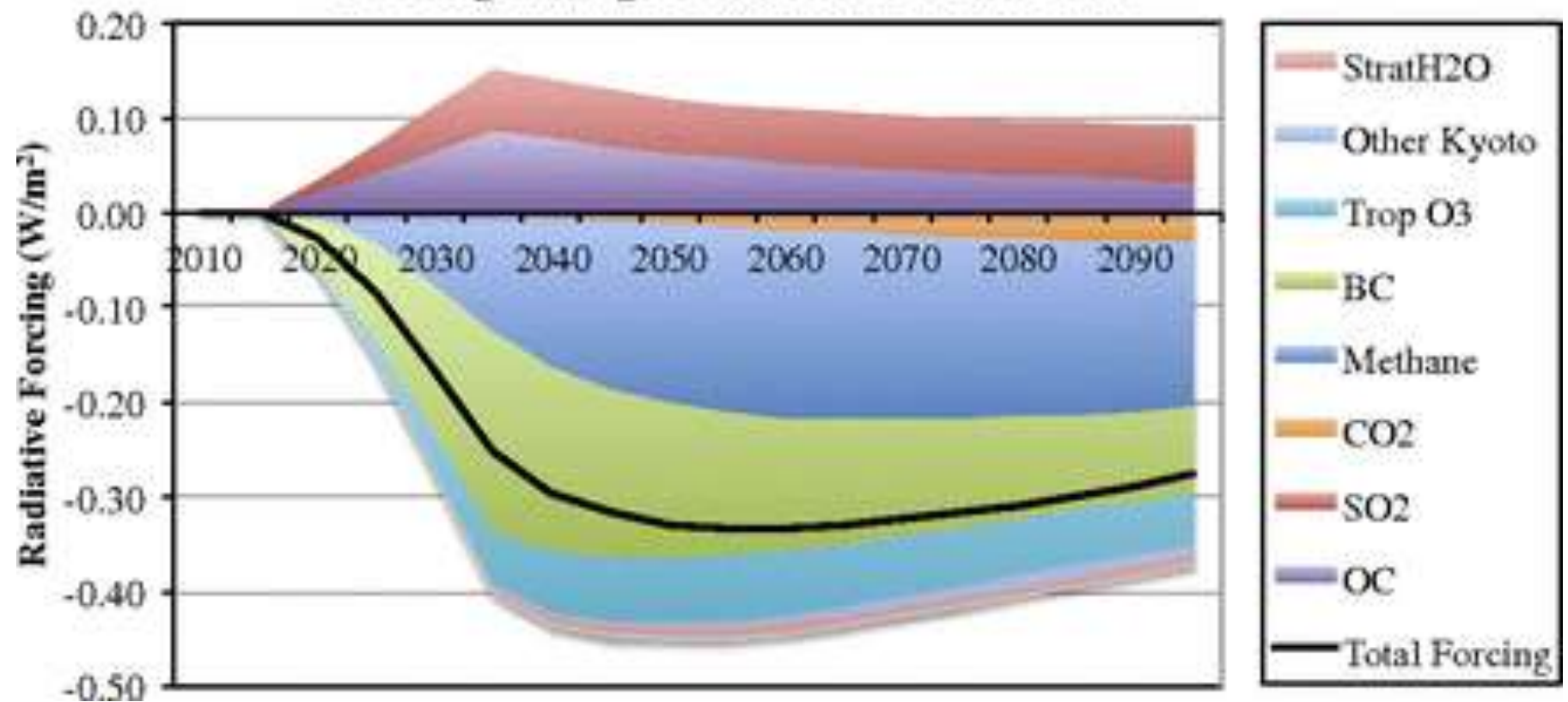
If our transport sector was shifted to EV and energy generation was completely shifted to renewables, tropospheric ozone mixing ratios would return to preindustrial levels within days to weeks.

In comparison: even if we could stop all CO_2 emissions right here and now, the CO_2 we emitted would still take almost a century to reduce to $1/e$ of the emitted amount.

Mitigation of ozone is financially attractive because the gas damages crops and is a pulmonary irritant.

People benefit where mitigation is pursued. Crop yields increase, health costs decrease. The beneficial effects of mitigation are local.

Forcing Change From SLCF Reduction



<https://doi.org/10.1073/pnas.1308470110>

BLACK CARBON (BC) MITIGATION

Black carbon, a climate forcer and air pollutant, is a product of the incomplete combustion of fossil fuels, biofuels and biomass.

The almost complete elimination of Kerosene lights in the past decade has been India's largest contribution to BC mitigation

Attempts to shift cooking to cleaner cook stoves aimed at improving health outcomes can result in equally large mitigation

Mitigating open crop residue burning and using the material better (e.g. for biochar soil amendment manufacturing) can result in mitigation + carbon capture and storage

The third largest source are diesel engines, which can be replaced by EV's (LDV's) and CNG

METHANE (CH₄) MITIGATION

Rice farming, cattle, sewage and biodegradable waste are the largest easy to mitigate anthropogenic methane sources.

Innovative rice farming techniques not only reduce the water footprint and input cost, they also reduce CH₄ emissions

Cow dung from large cattle herds can be profitably converted to bio CNG or electricity and can contribute a substantial share of the income earned. Biogas from large scale gober gas contributes around 9% of Germanys electricity supply.

Sewage treatment sludge and biodegradable waste can be used in biogas operations as well. However, this is easier to implement for sewage treatment sludge and industrial food waste generators (e.g. food processing industries, canteens) than for household waste (typically contaminated with plastic and other waste).

Biodegradable household waste can be put into a household scale biogas plant but the mitigation or damage potential which is determined by the methane leakage rate of such systems requires investigation. Aerobic composting of biodegradable waste and cooking on induction stoves powered by renewable electricity may be more beneficial to climate than small leaky biogas plants.

TROPOSPHERIC OZONE MIXING RATIOS

The transport sector, energy generation and industries are the three largest NO_x generators in India.

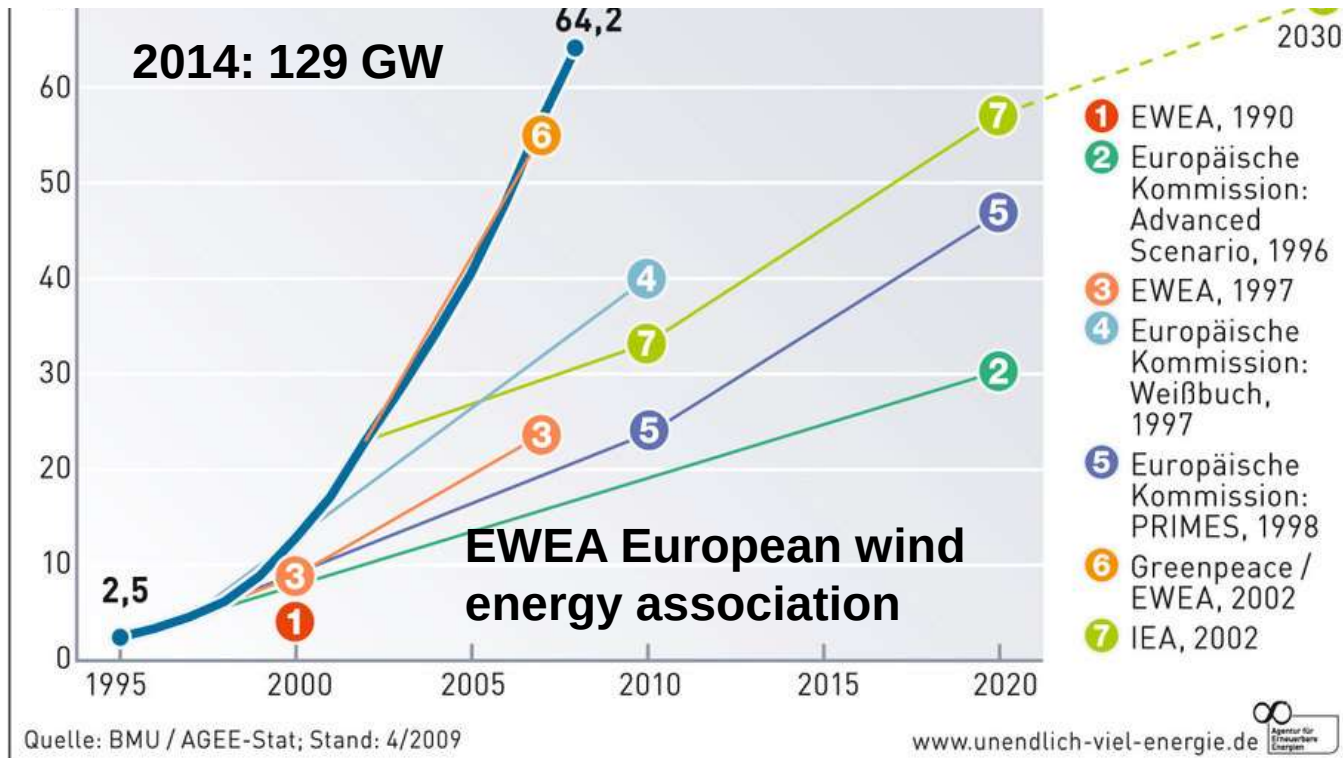
2-wheelers and 3-wheelers could easily be phased out in favour of EV's which are much cheaper to run ($1/6^{\text{th}}$ of the running cost) and do not depend on charging infrastructure (portable batteries). Freight could be shifted to rail if only the construction of the dedicated freight corridors would get finished ($1/5^{\text{th}}$ of the per km cost compared to truck).

Power generation from renewables is cheaper than from coal

Reducing the import of oil and coal could reduce the trade deficit.

Prognosis and reality

While wind power was becoming big and cheap, several agencies published estimates about how much it would contribute to the European electricity supply till 2020. The bold blue curve shows the **real development till 2008**. Note that the European Commission and the international Energy Agency never came close to capturing the real trend even for projections published as late as 2002. **Orders for wind parks that started feeding the grid in 2005-2006 HAD ALREADY BEEN PLACED BY THEN! Only the Greenpeace funded study captured the actual trend** but their projection was discredited as biased, although it was simply based on published order books from stock-market reporting major wind mill manufacturers!



If you want real news about what is going on with respect to renewables and the energy transition read “Business Insider”, “Forbes”, and quarterly stock market reports of major companies